Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/25/2013
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Investigation of Host-Pathogen Interactions, Conserved Cellular Responses, and Countermeasure Efficacy During Spaceflight using the Human Surrogate Model Caenorhabditis elegans		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Immunology		
Space Biology Special Category:	None		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	07/13/2010	**End Date:**	02/28/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 2/28/2015 per NSSC information (Ed., 3/13/14) NOTE: End date is now 7/12/2014 per NSSC information (Ed., 2/1/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alverdy, John (University of Chicago) Ott, C Mark (NASA Johnson Space Center)		
Grant/Contract No.:	NNX10AO52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Changes that occur to both the host immune system and pathogenesis of microbes during spaceflight represent a formidable challenge to the successful transition from short-to-long-duration spaceflight. This is a critical concern, since a) the immune system of the crew is dysfunctional during spaceflight, b) spaceflight uniquely increases the virulence and globally alters gene expression of the human pathogen Salmonella typhimurium when the bacteria are recovered post-flight and subsequently used to infect ground-based animal models, c) spaceflight-induced increases in S. typhimurium virulence are regulated by media ion composition, d) phosphate ion is sufficient to alter related pathogenesis responses in a ground-based spaceflight analogue model, and e) the evolutionarily conserved bacterial RNA chaperone protein, Hfq, serves as a master molecular regulator of many of these responses. While spaceflight has been shown to induce changes that can independently affect the host or the pathogen in a manner that is directly relevant to the development of infectious disease during spaceflight - none of these studies have been done when both the host and pathogen are simultaneously exposed to the spaceflight environment. To address this knowledge gap, we propose to challenge the human surrogate model Caenorhabditis elegans with S. typhimurium during spaceflight to study the role of the intestine in host-pathogen interactions in real-time, define virulence mechanisms, identify evolutionarily conserved responses, and test novel therapeutic strategies to prevent infectious disease. We hypothesize that the combination of a spaceflight-induced increase in virulence of the pathogen coupled with a blunted immune response of the host will result in a synergistic effect on the host-pathogen interaction such that the risk of infectious disease during spaceflight is exacerbated. We further propose that these responses represent evolutionarily conserved mechanisms at the level of both host and pathogen that are dependent on media ion composition, specifically the local concentration of phosphate. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	Previously published results from our lab showed that spaceflight uniquely increases the virulence and globally alters gene expression of the major human pathogen Salmonella typhimurium when the bacteria are recovered post-flight and subsequently used to infect laboratory-based animal models. In our efforts to determine the mechanism(s) behind the spaceflight-induced responses in Salmonella, we discovered that a) spaceflight-induced increases in Salmonella virulence are regulated by media ion composition, b) phosphate ion is sufficient to alter related pathogenesis responses in a spaceflight analogue model, c) the Hfq protein serves as a master molecular regulator of many of these responses in Salmonella, and d) Hfq also acts as a spaceflight-induced regulator acting across bacterial species, which to our knowledge, is the first report of such a master response regulator of the spaceflight-induced stimulon in bacteria. Hfq is a highly conserved bacterial RNA chaperone protein that binds to small non-coding regulatory RNAs thereby facilitating their association with mRNAs, the result of which plays a diverse role in prokaryotic gene expression (including regulation of ion response pathways), virulence, and physiology in response to stress. While spaceflight has been shown to induce changes that can independently affect the host or the pathogen in a manner that is directly relevant to the development of infectious disease during spaceflight – no studies have been published that document the outcome when both the host and pathogen are simultaneously exposed to the spaceflight environment. Furthermore, the eukaryotic organism and human surrogate infection model, C. elegans, contains structural and functional homologues of Hfq proteins, which are a family of evolutionarily conserved RNA-binding proteins that are critical for immune function, cell differentiation/function, and responses to cellular stress. We hypothesize that the combination of a spaceflight-induced increase in virulence of the pathogen coupled with a blunted immune response of the host will result in a synergistic effect on the host-pathogen interaction such that the risk of infectious disease during spaceflight is exacerbated and may uncover common conserved responses in the host and pathogen that can be targeted for countermeasure development. We have made significant progress toward achieving the objectives to successfully support a spaceflight investigation to meet a current SpaceX-3 launch date of Dec 9, 2013, including: - Continuing to work closely with BioServe flight hardware team to optimize hardware design and development to support this work on orbit – this includes obtaining funds to supplement BioServe's modification of video capabilities on orbit in order to provide required resolution of infected nematodes. - Developed and optimized protocols for culturing nematodes in liquid media. - Developed and optimized protocols for nematode purification. - Developed and optimized protocols to ensure long-term viability of nematodes in stasis media prior to on orbit activation. - Optimized nematode activation and infection parameters (including use of ground-based microgravity analogue culture systems - RWV bioreactors). - Optimized RNA stabilization and isolation parameters required for high integrity gene expression studies. - Optimizing fluorescent imaging and fixation techniques to visualize internalized bacteria. - Optimizing video capabilities for real time imaging of nematodes on orbit. - Performed a manifold test in our laboratory at ASU in conjunction with BioServe support to establish a preliminary baseline for hardware functionality.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	De Weirdt R, Crabbe A, Roos S, Vollenweider S, Lacroix C, van Pijkeren JP, Britton RA, Sarker S, Van de Wiele T, Nickerson CA. "Glycerol supplementation enhances L. reuteri's protective effect against S. Typhimurium colonization in a 3-D model of colonic epithelium." PLoS One. 2012;7(5):e37116. Epub 2012 May 31. PMID: 22693569 , Jun-2012

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Significant Media Coverage	Public Affairs Office, NASA Johnson Space Center. "Station Researchers Report Expanding Science Gains. Online article reporting on findings at AAAS meeting and PI quoted." March 8, 2013. http://www.nasa.gov/mission_pages/station/research/news/aaas.html , Mar-2013
Significant Media Coverage	Coulombe M, Derra S, Harth R, Hooker D, McBeath R, Snyder J. "Researchers talk science and society at AAAS national meeting. PI quoted." ASU News [science & tech]. February 20, 2013. https://asunews.asu.edu/20130220_aaasmeeting , Feb-2013
Significant Media Coverage	O'Brien M. "Spaceflight research highlighted on the PBS News Hour television documentary series with science correspondent Miles O’Brien, 2013." PBS News Hour television documentary series with science correspondent Miles O’Brien, 2013., Feb-2013